Finding Affordable Alternatives: A Similarity-Based System for Beauty Products

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Introduction

Background

- The beauty industry contains a wide range of products with significant price variations despite similar ingredient compositions and purposes.
- Consumers often seek lower priced alternatives that provide similar functionality to their higher-end counterparts.
- Identifying alternatives can be difficult due to inconsistent ingredient naming and lack of structured similarity tools.

Purpose

- The purpose of this project is to build a content-recommender system that recommends similar beauty products based on the ingredients. This system combines multiple natural language processing techniques to improve accuracy and consistency.

Discussion

- The hybrid approach improves recommendation quality by combining lexical, semantic, and set-based similarity measures. While TF-IDF captures exact ingredient matches, embeddings allow the model to recognize semantically similar products. Jaccard similarity reinforces structural overlap in ingredient lists.
- However, the model does not account for ingredient concentrations or formulation differences, which may impact real-world effectiveness.

Results

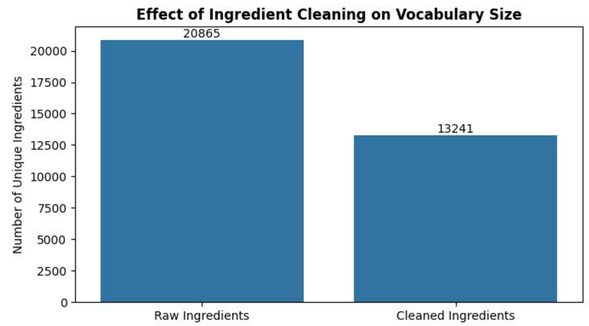

Figure 1: A graph showing how preprocessing/cleaning reduced number of unique ingredients in the set

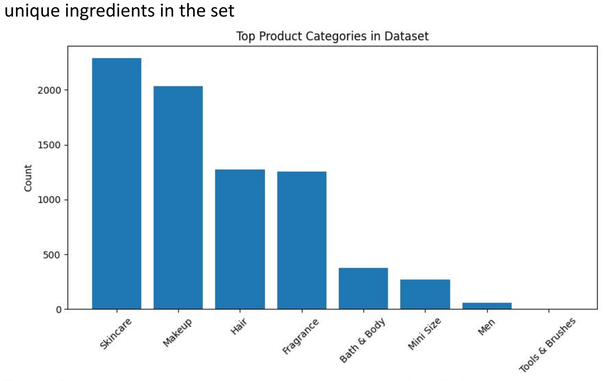

Figure 2: A graph showing the top product categories. This affects recommendations as recommendations are more limited for certain categories.

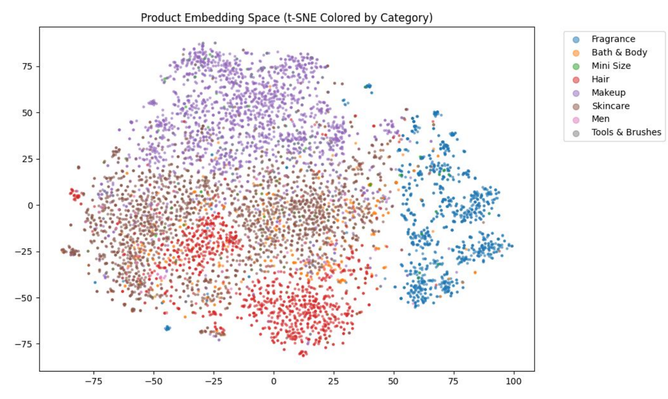

Figure 3: t-SNE highlights local similarity, producing distinct clusters for Fragrance, Makeup, and Hair, while showing more variation within Skincare and Bath & Body

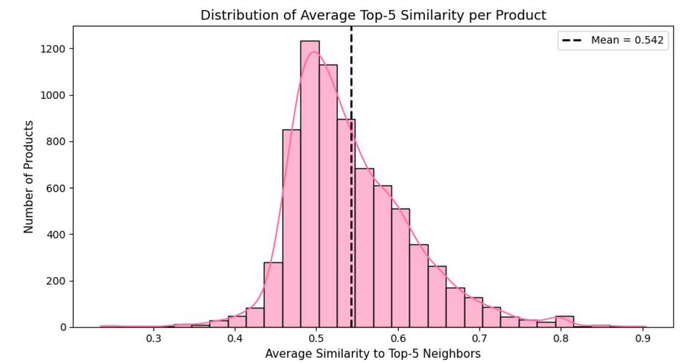

Figure 4: A graph showing distribution average of the top 5 products to see model performance

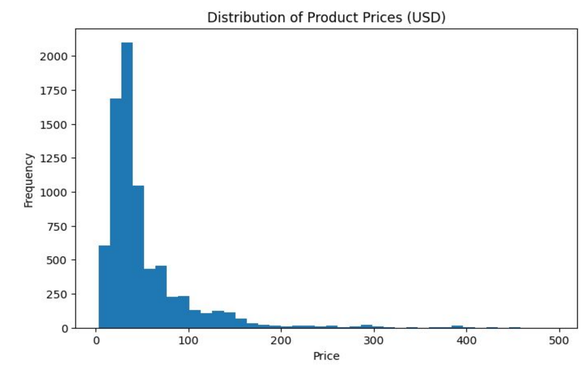

Figure 5: A graph showing the distribution of product prices to show that many of the product recommendations are mid-tier reflecting sephora's general prices.

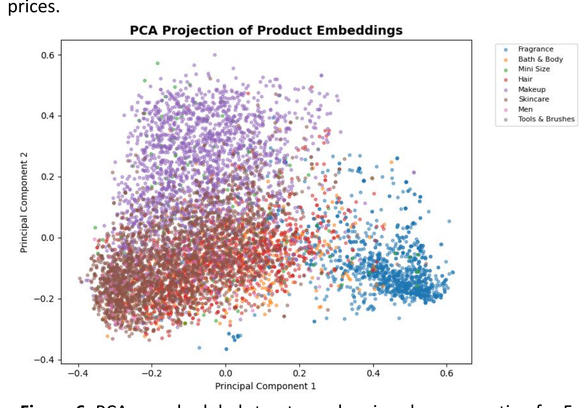

Figure 6: PCA reveals global structure, showing clear separation for Fragrance but significant overlap among Skincare, Makeup, and Hair, indicating shared product characteristics.

Method

- The dataset is from kaggle. It consists of web scraped data from Sephora's API that includes features such as product name, cost, ingredients, reviews, highlights, brand, category, and more.
- The data was preprocessed to standardize and reduce noise. This included mapping synonymous terms.
- TF-IDF captured keyword importance and ingredient frequency. Sentence-BERT embeddings captured semantic meaning from ingredients and product descriptions. Jaccard similarity measured overlap between ingredient sets.
- A similarity score was computed as a weighted combination of the three methods above. Recommendations are generated by ranking products based on similarity while ensuring it is in the same category and are lower in price.

References

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Conclusion

- This project demonstrates that hybrid NLP techniques can effectively identify lower-cost alternatives in the beauty domain. The system provides consistent, high-similarity recommendations while maintaining price constraints, making it a practical tool for consumer decision support.
- Future improvements include incorporating ingredient concentrations, using user reviews, and expanding the dataset.